

INDIAN STATISTICAL INSTITUTE
Assignment-6 (Mathematics III)
Bachelor of Statistical Data Science (BSDS)

- (1) Show that the norm $\|x\|_1 = \sum_{n=1}^k |x_n|$ on the space $\mathbb{R}^k$ is not induced by an inner product.
(2) If $X = \mathbb{R}^k$ and $A = \{a\}$, for some non-zero vector $a = (a_1, \dots, a_k)$, show that

$$A^\perp = \{(x_1, x_2, \dots, x_k) \in \mathbb{R}^k : \sum_{j=1}^k a_j x_j = 0\}.$$

- (3) If $A = \{\{x_n\} \in l^2 : x_{2n} = 0 \text{ for all } n \in \mathbb{N}\}$, find $A^\perp$.
(4) Let $\mathcal{H}$ be a Hilbert space, let $y \in \mathcal{H} \setminus \{0\}$ and let $S = \text{Span}\{y\}$. Show that

$$\{x \in \mathcal{H} : \langle x, y \rangle = 0\}^\perp = S.$$

- (5) Any infinite-dimensional inner product space X contains an orthonormal sequence.
(6) Let $\mathcal{H}$ be a Hilbert space and let $\{e_n\}$ be an orthonormal basis in $\mathcal{H}$. Prove that the Parseval relation

$$\langle x, y \rangle = \sum_{n=1}^{\infty} \langle x, e_n \rangle \langle e_n, y \rangle.$$

- (7) Let $\mathcal{H}$ be a Hilbert space and let $\{e_n\}$ be an orthonormal basis in $\mathcal{H}$. Determine whether the following series converge in $\mathcal{H}$:

$$a) \quad \sum_{n=1}^{\infty} \frac{e_n}{n}, \quad b) \quad \sum_{n=1}^{\infty} \frac{e_n}{n^{\frac{1}{2}}}.$$